

ADDITIVE MANUFACTURE, 3D-PRINTED, METALLIC X-RAY MIRROR FOR SYNCHROTRON AND XFEL FACILITIES

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Abstract

We have designed and fabricated the world's first, Additive Manufacture (AM) mirror for X-ray beamlines. For traditional optics, beamline performance is limited by: distortion caused by mechanical clamping; heat-bumps induced by photon-beam illumination; and strain caused by differential thermal expansion when dissimilar materials are cooled. AM enables the creation of intricate internal structures, and the fusion of multiple components into a single piece. The optical substrate, beamline mount, and internal cooling manifold were combined into a monolithic structure. The X-ray mirror was 3D-printed in aluminium alloy AlSi10Mg. Single-point diamond turning created an optical surface, which was coated in $\sim 75\text{ }\mu\text{m}$ of electroless NiP (nickel phosphor), followed by "super-polishing" using chemo-mechanical processing. Optical metrology demonstrates the AM mirror has surface quality comparable to a traditional silicon mirror. The design is virtually immune to clamping deformations, which simplifies beamline installation. AM unlocks exotic internal channel designs, including enhanced cooling performance by turbulent flow, reducing vibrations caused by fluid flow, and conforming to the heat load distribution.

INTRODUCTION

Additive Manufacture is increasingly used for creating novel engineering components and lightweighted optics for space-based astronomy [1] that would be challenging or impossible to create using traditional subtractive methods. We extend this concept to creating a composite X-ray mirror system for synchrotron light and XFEL beamlines.

MANUFACTURING

After design by the Engineering team at Diamond Light Source, based on thermal and mechanical CAD modelling, the mirror was grown using Additive Manufacture (AM) at CA Models, UK (see Fig. 1). This was followed by stress-relieving at 300°C for 2 hours. AlSi10Mg was selected as a cost-effective, and well understood aluminium alloys routinely used for additive manufacturing. One limitation of this alloy is its relatively high thermal expansion coefficient ($20 \times 10^{-6}\text{ K}^{-1}$). AlSi40Mg promises an improved thermal expansion coefficient of $13 \times 10^{-6}\text{ K}^{-1}$, which would ease the management of thermal distortion for future optics. However, AM manufacture using this

alloy is not-readily commercially available and is still in the materials research phase [2].

After growth and annealing, the surface was diamond-turned at Durham University, UK to create a basic optical finish. It was then coated with a $75\text{ }\mu\text{m}$ layer of NiP since aluminium is unsuitable for super-polishing due to its lower hardness and polycrystalline structure. After coating, the surface quality would be sufficient for infra-red optics, but not for the more demanding specifications for X-ray applications at synchrotron or XFEL facilities.

Super-polishing at Crystal Scientific, UK, improved the micro-roughness and form. Finally, ion beam figuring (IBF) at Diamond [3], reduced the height and slope errors (see Fig. 2).

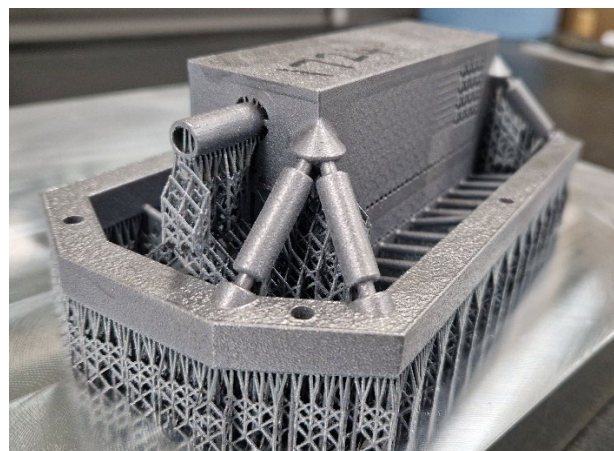


Figure 1: 3D-printed, additive manufactured, X-ray mirror for synchrotron beamlines, combines a high-quality optical surface with novel internal cooling channels and a triple-bipod clamping frame to make beamline installation easier.

METROLOGY

A Bruker GTX micro-interferometer and Zygo HDX Fizeau interferometer were used in the Optics Metrology Lab (OML) at Diamond [4] to measure the micro-roughness, height and slope errors of the mirror at various stages during production (Fig. 3). Metrology data was also used to guide deterministic correction of surface errors using IBF. A further thermal study was undertaken in the OML to observe how the optical surface changes as water was flowed at various temperatures and rates through the internal cooling channels using a portable chiller and pump (Fig. 4).

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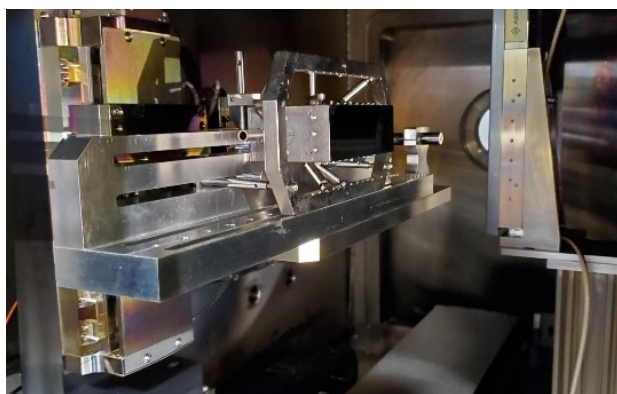


Figure 2: Ion beam figuring at Diamond Light Source to improve the surface form and figure errors of the AM mirror.

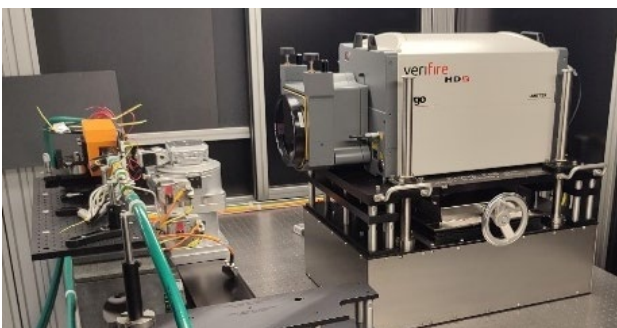


Figure 3: Fizeau interferometry to assess the surface quality of the finished mirror and how the optical surface changed as cooling water was flowed through the internal cooling circuit.

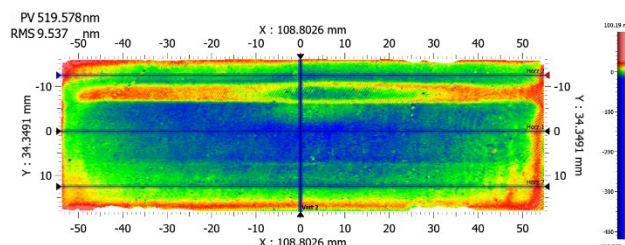


Figure 4: Height map of the optical surface, measured by Fizeau interferometry, showing changes as cooling water was flowed through the internal cooling channels.

CONCLUSIONS

This prototype study produced an X-ray mirror (Fig. 5) using Additive Manufacturing with suitable performance for operation on a synchrotron or XFEL beamline. All design criteria were achieved. Lessons learned during the production stage will be applied for future projects.

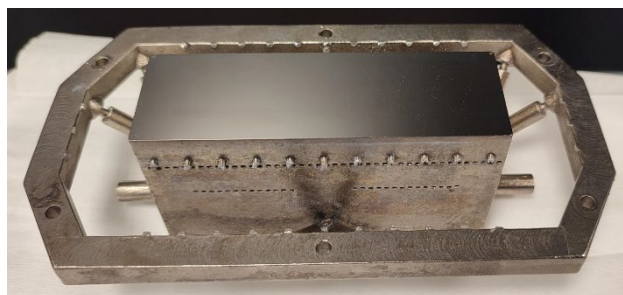


Figure 5: Photograph of the final X-ray mirror which has optical properties suitable for operation on synchrotron beamlines.

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